

input design and core qualitative methods as core design (quan-QUAL), (ii) supplementary quantitative methods design as follow up to core quantitative methods design (QUAL-quan), (iii) preliminary qualitative methods input design and core quantitative methods design (qual-QUAN), (iv) supplementary qualitative methods design as follow up to quantitative methods design (QUAN-qual).

6.10 SOME USEFUL BOOKS

Bergman Manfred Max (2008): *The Strawmen of Qualitative-Quantitative Divide and Their Influence on Mixed Method Research in Advances in Mixed Methods Research*, edited by Manfred Max Bergman, Sage Publications Ltd. London.

Creswell John W, Clark Kicki L Plano and Amanda L. Garrett (2008): *Methodological Issues in Conducting Mixed Methods Research Designs in Advances in Mixed Methods Research*, Sage Publications Ltd. London.

Tashakkori Abbas and Teddlie Charles (2008): *Quality of inferences in Mixed Methods Research in Advances in Mixed Methods Research*, Sage Publications Ltd. London.

Morgan David L (2014): *Integrating Qualitative and Quantitative Methods- A Pragmatic Approach*, Sage Publications, London, N. Delhi, Singapore.

6.11 ANSWERS OF HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) A variable expressed in terms of numerals like 1, 2, 3,.. etc. is called quantitative variable. For instance, yield of an agricultural output, height/weight of individuals, etc. can be measured and expressed in quantitative terms. On the other hand, a variable expressed in terms of the choices or opinion about attributes/ characteristics like one's feelings or opinion is called qualitative variable.
- 2) See Sub-section 6.2.1
- 3) See Sub-section 6.2.3
- 4) Descriptive research describes a situation, a phenomenon or event or a social system. It aims to describe the state of affairs as it exists. On the other hand, explanatory results aim to establish cause and effect relationship.

Check Your Progress 2

- 1) Research questions guide the research design.
- 2) There are two major functions of the research design: (i) to determine the type of evidences (data/information) which are needed to answer the research questions, (ii) to eliminate the alternative explanation of the results.

- 3) If a research study is confined to use **either** quantitative **or** qualitative method, it is said to be a mono method research. The combination of at least one quantitative and at least one qualitative component in a single research project is known as mixed methods research.

Check Your Progress 3

- 1) See Section 6.6
- 2) See Section 6.7
- 3) See Section 6.7



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UNIT 7 DATA COLLECTION AND SAMPLING DESIGN

Structure

- 7.0 Objectives
- 7.1 Introduction
- 7.2 An Overview of the Unit
- 7.3 Method of Data Collection
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- 7.6 The Choice of an Appropriate Sampling Method
- 7.7 Let Us Sum Up
- 7.8 Some Useful Books
- 7.9 Answers or Hints to Check Your Progress Exercises

7.0 OBJECTIVES

After going through this Unit, you will be able to:

- appreciate different methods of collecting data;
- acquire knowledge of different tools of data collection;
- define the key terms commonly used in quantitative analysis, like parameter, statistic, estimator, estimate, inference, standard error, confidence intervals, etc.;
- distinguish between random and non-random sampling procedures for data collection;
- appreciate the advantages of random sampling in the assessment of the “precision” of the estimates of population parameters;
- acquire knowledge of the procedure for drawing samples by different methods;

- develop the ability to obtain estimates of key parameters like population and sample aggregates, proportion, mean, etc.; and of the “precision” of such estimates under different sampling methods; and
- appreciate the feasibility/appropriateness of applying different sampling methods in different research contexts.

7.1 INTRODUCTION

Studies of the behaviour of variables and of relationships amongst them necessitate measurement of the variables involved. Variables can be *quantitative variables* like GNP or *qualitative variables* like opinions of individuals on, say, ban on smoking in public places. The former set assumes quantitative values. The latter set does not admit of easy quantification, though some of these can be categorised into groups that can then be assigned quantitative values. Research strategies thus adopt two approaches, quantitative and qualitative. We shall deal with the quantitative approach in this unit. The various measurement scales and scaling techniques to measure the qualitative data will be discussed in Unit 8.

The basic ingredient of quantitative research is the measurement in quantitative terms of the variables involved or the collection of the data relevant for the analytical and interpretative processes that constitute research. The quality of the data utilised in research is important because the use of faulty data in such endeavour results in misleading conclusions, however sophisticated may be the analytical tools used for analysis. Research processes, be it testing of hypotheses and models or providing the theoretical basis for policy or review of policy, call for objectivity, integrity and analytical rigour in order to ensure academic and professional acceptability and, above all, an effective tool to tackle the problem at hand. Data used for research should, therefore, reflect, as accurately as possible, the phenomena these seek to measure and be free from errors, bias and subjectivity. Collection of data has thus to be made on a scientific basis.

7.2 AN OVERVIEW OF THE UNIT

How to assemble data on a scientific basis? There are broadly three different methods of collecting data. These are dealt with in section 7.3. The tools that one can use for collecting data – the formats and the devices that modern technology has provided – are enumerated in section 7.4. There are situations where it is considered desirable to gather data from only *a part* of the universe, or a *sample* selected from the universe of interest to the study at hand, rather than a complete coverage of the universe, for reasons of cost, convenience, expediency, speed and effort. Questions then arise as to the manner in which such a *sample* should be chosen – the sampling design. This question is examined in detail in section 7.5. The discussion is divided into a number of sub-topics. Concepts relating to population aggregates like mean and variance and similar aggregates from the sample and the use of the latter as estimates of population aggregates have been introduced in sub-section 7.5.1. There are two types of sampling: random and non random. Non random sampling methods and the contexts in which these are used are described in sub-section 7.5.2. A random sample has certain advantages over a non random sample — it provides a basis for drawing *valid* conclusions from the sample about the parent population. It enables us to state the precision of the estimates of

population parameters in terms of (a) the extent of their variation or (b) an interval within which the value of the population parameter is likely to lie with a given degree of certainty. Further, it even helps the researcher to determine the size of the sample to be drawn if his project is subject to a sanctioned budget and permissible limits of error in the estimate of the population parameter. These principles are explained in sub-section 7.5.3. Eight methods of random sampling are then detailed in sub-section 7.5.4. These details relate to (i) operational procedures for drawing samples, and (ii) expressions for (a) estimators of parameters and measures of their variation and b) estimators of such variation where the population variation parameter is not known. Different sampling procedures are also compared, as we go along, in terms of the relative precision of the estimates, they generate. Finally, the question of choosing the sampling method that is appropriate to a given research context is addressed in section 7.6. A short summing up of the Unit is given in section 7.7.

7.3 METHOD OF DATA COLLECTION

There are three methods of data collection – *the Census and Survey Method*, *the Observation Method* and *the Experimental Method*. The first is a carefully planned and organised study or enquiry to collect data on the subject of the study/enquiry. We might for instance organise a study on the prevalence of the smoking habit among high school children – those aged 14 to 17 - in a certain city. One approach is to collect data of the kind we wish to collect on the subject matter of the study from *all* such children in *all* the schools in the city. In other words, we have a **complete enumeration** or **census** of the **population** or **universe** relevant to the enquiry, namely, the city's high school children (called the *respondent units* or *informants* of the Study) to collect the data we desire. The other is to confine our attention to a *suitably selected part of the population* of high school children of the city, or a **sample**, for gathering the data needed. We are then conducting a *sample survey*. A well known example of Census enquiry is the **Census of Population** conducted in the year **2011**, where data on the demographic, economic, social and cultural characteristics of *all persons* residing in India were collected. Among **sample surveys** of note are the household surveys conducted by the National Sample Survey Organisation (NSSO) of the Government of India that collect data on the socio-economic characteristics of a sample of households spread across the country.

The Observation Method records data as things occur, making use of an appropriate and accepted method of measurement. An example is to record the body temperature of a patient every hour or a patient's blood pressure, pulse rate, blood sugar levels or the lipid profile at specified intervals. Other examples are the daily recording of a location's maximum and minimum temperatures, rainfall during the South West / North East monsoon every year in an area, etc.

The Experimental Method collects data through *well designed and controlled* statistical experiments. Suppose for example, we wish to know the rate at which manure is to be applied to crops to maximise yield. This calls for an experiment, in which all variables other than manure that affect yield, like water, quality of soil, quality of seed, use of insecticides and so on, need to be controlled so as to evaluate the effect of different levels of manure on the yield. Other methods of conducting the experiment to achieve the same objective without controlling "all other factors" also exist. Two branches of statistics – The Design and Analysis of Experiments and Analysis of Variance — deal with these.

7.4 TOOLS OF DATA COLLECTION

How do we collect data? We translate the data requirements of the proposed Study into items of information to be collected from the respondent units to be covered by the study and organise the items into a logical format. Such a format, setting out the items of information to be collected from the respondent units, is called *the questionnaire or schedule* of the study. The questionnaire has a set of pre-specified questions and the replies to these are recorded either by the respondents themselves or by the investigators. The *questionnaire approach* assumes that the respondent is capable of understanding and answering the questions all by himself/herself, as the investigator is not supposed, in this approach, to influence the response in any manner by interpreting the terms used in the questions. Respondent-bias will have to be minimised by keeping the questions simple and direct. Often the responses are sought in the form of “yes”, “no” or “can’t say” or the judgment of the respondent with reference to the perceived quality of a service is graded, like, “good”, “satisfactory” or “unsatisfactory”.

In the *schedule approach* on the other hand, the questions are detailed. *The exact form of the question to be asked of the respondent is not given to the respondent and the task of asking and eliciting the information required in the schedule is left to the investigator.* Backed by his training and the instructions given to him, the investigator uses his ingenuity in explaining the concepts and definitions to respondents to obtain reliable information. This does not mean that investigator-bias is more in the schedule approach than in the questionnaire approach. Intensive training of investigators is necessary to ensure that such a bias does not affect the responses from respondents.

Schedules and questionnaires are used for collecting data in a number of ways. Data may be collected by personally contacting the respondents of the survey. Interviews can also be conducted over the telephone and the responses of the respondent recorded by the investigator. The advent of modern electronic and telecommunications technology enables interviews being done through e-mails or by ‘chatting’ over the internet. The mail method is one where (usually) questionnaires are mailed to the respondents of the survey and replies received by mail through (postage pre-paid) business-reply envelopes. The respondents can also be asked (usually by radio or television channels or even print media) to send their replies by SMS to a mobile telephone number or to an e-mail address.

Collection of data can also be done through mechanical, electro-mechanical or electronic devices. Data on arrival and departure times of workers are obtained through a mechanical device. The time taken by a product to roll off the assembly line and the time taken by it to pass through different work stations are recorded by timers. A large number of instruments are used for collecting data on weather conditions by meteorological centres across the country that help assessing current and emerging weather conditions. Electronic Data Transfers (EDT) can also be the means through which source agencies like ports and customs houses, where export and import data originate, supply data to a central agency like the Directorate General of Commercial Intelligence and Statistics (DGCI&S) for consolidation.

The above methods enable us to collect *primary data*, that is, data being *collected afresh* by the agency conducting the enquiry or study. . The agency

concerned can also make use of data on the subject *already collected* by another agency or other agencies – *secondary data*. Secondary data are published by several agencies, mostly Government agencies, at regular intervals. These can be collected from the publications / compact discs or the websites of the agencies concerned. But such data have to be examined carefully to see whether these are suitable or not for the study at hand before deciding to collect new data.

Errors in data constitute an important area of concern to data users. Errors can arise due to confining data collection to a sample. (*sampling errors*). It can be due to faulty measurement arising out of lack of clarity about what is to be measured and how it is measured. Even when these are clear, errors can creep in due to inaccurate measurement. Investigator bias also leads to errors in data. Failure to collect data from respondent units of the population or the sample due to omission by the investigator or due to non-response (respondents not furnishing the required information) also results in errors. (*non-sampling errors*). The *total survey error* made up of these two types of errors need to be minimised to ensure quality of data.

7.5 SAMPLING DESIGN

We have looked at methods and tools of data collection, chief among which is the sample survey. How to select a sample for the survey to be conducted? There are a number of methods of choosing a sample from a universe. These consist of two categories, *random sampling* and *non-random sampling*. Let us turn these methods and see how well the results from the sample can be utilised to draw conclusions about the parent universe.

But first let us turn to some notations, concepts and definitions.

7.5.1 Population and Sample Aggregates and Inference

Let us denote population characteristics by upper case (capital) letters in English or Greek and sample characteristics by lower case (small) letters in English. Let us consider a (finite) population consisting of N units U_i ($i = 1, 2, \dots, N$). Let Y_i ($i = 1, 2, \dots, N$) be the value of the variable y , the characteristic under study, for the i^{th} unit U_i ($i = 1, 2, \dots, N$). For instance, the units may be the students of a university and y may be their weight in kilograms. Any function of the population values Y_i is called a **parameter**. An example is the population mean ' μ ' or ' M ' given by $(1/N) \sum_{i=1}^N Y_i$, where $\sum_{i=1}^N$ stands for summation over $i = 1$ to N . Let us now draw a sample of ' n ' units u_i ($i = 1, 2, \dots, n$)¹ from the above population and let the value of the i^{th} sample unit be y_i ($i = 1, 2, \dots, n$)². In other words, y_i ($i = 1, 2, \dots, n$) are the sample observations. A function of the sample observations is referred to as a **statistic**. The sample mean ' m ' given by $(1/n) \sum_{i=1}^n y_i$, $\sum_{i=1}^n$ ($i = 1$ to n), is an example of a statistic.

Let us note the formulae for some important parameters and statistics.

$$\text{Population total } Y = \sum_{i=1}^N Y_i, \quad \sum_{i=1}^N \text{ stands for summation over } i = 1 \text{ to } N \quad (2.1)$$

$$\text{Population mean } = \mu \text{ or } M, = (1/N) \sum_{i=1}^N Y_i, \quad \sum_{i=1}^N, i = 1 \text{ to } N \quad (2.2)$$

¹ The sample units are being referred to as u_i ($i = 1, 2, \dots, n$) and not in terms of U_i as we do not know which of the population units have got included in the sample. Each u_i in the sample is some population unit

² The same reasons apply for referring the sample values or observations as y_i ($i = 1, 2, \dots, n$) and not in terms of the population values Y_i . y_i will be some Y_i .

$$\text{Population variance } \sigma^2 = (1/N) \sum_{i=1}^N Y_i^2 - M^2, \quad i, i = 1 \text{ to } N \quad (2.3)$$

$$\text{Population SD } = \sigma = +\sqrt{[(1/N) \sum_{i=1}^N Y_i^2 - M^2]}, \quad i, i = 1 \text{ to } N \quad (2.4)$$

$$\text{Sample mean } = (1/n) \sum_{i=1}^n y_i, \quad i, (i = 1 \text{ to } n) \quad (2.5)$$

$$\text{Sample variance } s^2 = (1/n) \sum_{i=1}^n y_i^2 - m^2, \quad i, i = 1 \text{ to } n \quad (2.6)$$

$$= [ss]^2 / n, \text{ where } [ss]^2 = \sum_{i=1}^n (y_i - m)^2 = \sum_{i=1}^n y_i^2 - n m^2 =$$

sum of squares of sample observations from their mean 'm' (2.7)

$$\text{Sample standard deviation 's'} = +\sqrt{[(1/n) \sum_{i=1}^n y_i^2 - m^2]}, \quad i, i = 1 \text{ to } n \quad (2.8)$$

$$\text{Population proportion } P = (1/N) \sum_{i=1}^N Y_i = N^1 / N, \text{ (where } N^1 \text{ is the number of units in the population possessing a specified characteristic)} \quad (2.9)$$

$$\sigma^2 = (1/N) \sum_{i=1}^N Y_i^2 - M^2 = P - P^2 = P(1 - P) = PQ,$$

$$\text{where } Q = [(N - N^1)/N] = (1 - P). \quad (2.10)$$

$$m = p, \text{ (proportion of units in the sample with the specific characteristic)} \quad (2.11)$$

$$s^2 = p(1 - p) = pq, \text{ where } p \text{ is the sample proportion and } p + q = 1 \quad (2.12)$$

$$[ss]^2 = npq \quad (2.13)$$

The purpose of drawing a sample from a population is to arrive at some conclusions about the parent population from the results of the sample. This process of drawing conclusions or making inferences about the population from the information contained in a sample chosen from the population is called **inference**. Let us see how this process works and what its components are. The sample mean 'm', for example, can serve as an estimate of the value of the population mean 'μ'. The statistic 'm' is called an **estimator (point estimator)** of the population mean 'μ'. The value of 'm' calculated from a specific sample is called an **estimate (point estimate)** of the population mean 'μ'. In general, *a function of sample observations, that is, a statistic, which can be used to estimate the unknown value of a population parameter, is an estimator of the population parameter. The value of the estimator calculated from a specific sample is an estimate of the population parameter.*

The estimate 'm₁' of the population parameter 'μ', computed from a sample, will most likely be different from 'μ'. There is thus an *error* in using 'm₁' as an estimate of 'μ'. This error is the sampling error, assuming that all measurement errors, biases etc., are absent, that is, *there are no non-sampling errors*. Let us draw another sample from the population and compute the estimate 'm₂' of 'μ'. 'm₂' may be different from 'm₁' and also from 'μ'. Supposing we generate in this manner a number of estimates m_i (i = 1,2,3,.....) of 'μ' by drawing repeated samples from the population. All these m_i (i = 1,2,3,....) would be different from each other and from 'μ'. What is the extent of the variability in the m_i (i = 1,2,3,....), or, the variability of the error in the estimate of 'μ' computed from different samples? How will these values be spread or scattered around the value of 'μ' or the errors be scattered around zero? What can we say about the estimate of the parameter obtained from the specific sample that we have drawn from the population as a means of measuring the parameter, without actually drawing repeated samples? How well do non-random and random samples answer these questions? The answers to these questions are important from the point of view of inference.

Let us first look at the different methods of non-random sampling and then move on to random sampling.

7.5.2 Non-Random Sampling

There are several kinds of non-random sampling. A **judgment sample** is a sample that has been selected by making use of one's expert knowledge of the population or the universe under consideration. It can be useful in some circumstances. An auditor for example could decide, on the basis of his experience, on what kind of transactions of an institution he would examine so as to draw conclusions about the quality of financial management of an institution. **Convenience Sampling** is used in exploratory research to get a broad idea of the characteristic under investigation. An example is one that consists of some of those coming out of a movie theatre; and these persons may be asked to give their opinion of the movie they had just seen. Another example is one consisting of those passers by in a shopping mall whom the investigator is able to meet. They may be asked to give their opinion on a certain television programme. The point here is the convenience of the researcher in choosing the sample. **Purposive Sampling** is much similar to judgement sampling and is also made use of in preliminary research. Such a sample is one that is made up of a group of people specially picked up for a given purpose. **In Quota Sampling**, subgroups or strata of the universe (and their shares in the universe) are identified. A **convenience** or a **judgement sample** is then selected from each stratum. No effort is made in these types of sampling to contact members of the universe who are difficult to reach. In Heterogeneity Sampling units are chosen to include all opinions or views. **Snowball Sampling** is used when dealing with a rare characteristic. In such cases, contacting respondent units would be difficult and costly. This method relies on referrals from initial respondents to generate additional respondents. This technique enables one to access social groups that are relatively invisible and vulnerable. This method can lower search costs substantially but this saving in cost is at the expense of the representative character of the sample. An example of this method of sampling is to find a rare genetic trait in a person and to start tracing his lineage to understand the origin, inheritance and etiology of the disease.

It would be evident from the description of the methods given above that the relationship between the sample and the parent universe not clear. The selection of specific units for inclusion in the sample seem to be *subjective* and *discretionary* in nature and, therefore, may well reflect the researcher's or the investigator's attitudes and bias with reference to the subject of the enquiry.

A sample has to be *representative* of the population from which it has been selected, if it is to be useful in arriving at conclusions about the parent population. A **representative sample** is one that contains the relevant characteristics of the population in the same proportion as in the population. Seen from this angle, the non-random sampling methods described above do not yield representative samples. Such samples are, therefore, not helpful in drawing *valid* conclusions about the parent population and *the way these conclusions change* when another sample is chosen from the population. Non-random sampling is, however, useful in certain circumstances. For instance, it is an inexpensive and quick way to get a preliminary idea of the variable under study or a rough preliminary estimate of the characteristics of the universe that helps us to design a scientific enquiry into the problem later. It is thus useful in exploratory research.

Check Your Progress 1

- 1) Name the three methods of the data collection

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- 2) What is meant by a 'parameter'? Defining the term 'statistic', indicate the expressions for population/sample mean and population/sample mean and population/sample variance.

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- 3) Mention the most important purpose of studying the population on the basis of a sample. In this context, define the terms 'estimator' and 'estimate' with a suitable example.

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- 4) Defining the term 'representative sample'. Indicate how is 'random sampling' principally different from that of 'non-random sampling'? What could be the use of the latter despite its major drawback vis-à-vis the former?

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7.5.3 Random or Probability Sampling

Random sampling methods, on the other hand, yield samples that are representative of the parent universe. The selection process in random sampling is free from the bias of the individuals involved in drawing the sample as the units of the population are selected at random for inclusion in the sample. *Random sampling is a method of sampling in which each unit in the population has a predetermined chance (probability) of being included in the sample. A sampling design is a clear specification of all possible samples of a given type with their corresponding probabilities.* This property of random sampling helps us to answer the questions we raised at the end of sub-section 2.6.1 above. That is, we can make estimates of the characteristics of the parent population from the results of a sample and also indicate the extent of error to which such estimates are subject or the *precision of the estimate*. This is better than not knowing anything at all about the magnitude of the error in our statements regarding the parent population. Let us see how random sampling helps in this regard.